

# Synthetic Benchmarking of Document Classifiers Under Domain Shift

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This document is fictional and exists only to demonstrate page-level evidence verification.

## Abstract

This fictional benchmark evaluates document classifiers under in-domain and held-out-domain conditions using a synthetic abstract corpus.

## Methods

The corpus contained 2,400 synthetic labeled abstracts across four domains. The primary grouped split held out one complete domain before model fitting.

A TF-IDF linear classifier and a compact transformer were evaluated using macro-F1 averaged across five fixed seeds.

A secondary random document split was included only as a leakage-sensitivity comparison.

## Results

Transformer macro-F1 was **0.86 in-domain** and **0.71 on the held-out domain**.

A random document split inflated transformer macro-F1 by **0.06** compared with the grouped domain-aware split.

| Model         | In-domain macro-F1 | Held-out macro-F1 |
|---------------|--------------------|-------------------|
| TF-IDF linear | 0.78               | 0.65              |
| Transformer   | 0.86               | 0.71              |

**Benchmark performance is meaningful only with the dataset, split, metric, and leakage boundary.**

## Limitations

The synthetic corpus and limited domain diversity restrict external validity. Only one synthetic held-out domain was evaluated.

The benchmark does not establish performance on full-text articles, multilingual corpora, or real deployment data.

## Data and license

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